

1. INTRODUCTION

1.1 History of Solar Pumps

The increasing global energy shortages and the pollution of natural habitats have been for decades attracted by the fraternity of engineering and science. Awareness for the useful production of renewable energies has rapidly matured. A significant contribution to energy conversion was the advent of rapid switching electronics and growth of semiconductor technology. The use of renewable energy, which began when fluvial water energy was transformed, has today directly transported solar energy into electricity.

PV converters were inefficiently effective at 5-6 percent and were extremely expensive in earlier times. PV converters have been highly inefficient. However, the performance of the PV array has at present reached 15-16%, increased research and development technologies. Moreover, concentrations have declined gradually. The conversion of photovoltaic energy is now considered to be one of the promising alternatives to fossil-fuel power systems as non dangerous waste, no greenhouse gas emissions, no fuel costs involved, low maintenance costs, water usage etc. The technology is however under development and many issues must be addressed, including intermittence, high start-up costs and poor performance.

Solar water pumps become more popular in rural areas, where electricity is not available. Solar PV pumps are preferred in remote areas for irrigation, water treatment and farming. For good yield in countries such as India, irrigation is important, which has 70% of its population depends on agriculture. In the planet, there are many water pumps with electricity or nonrenewable energy sources. In terms of costs and emissions, the acquisition of solar-photovoltaic pumping systems. Better than diesel water pumping systems is better.

In order to satisfy the needs of operations with variable power constraints, a photo voltaic powered driving system requires a number of innovative solutions. An induction motor drive for PVP applications(IMD) shows good performance in its robust nature in contrast to other commercial motors. The goal of the evolution is to develop an effective, reliable PV water pumping system that is free and affordable for maintenance. New permanent magnet engines like a brushless DC engine and a permanent magnet sinus engine are used for pumping but, as a result of cost and availability limitations, still overshadowed by induction motors.

Manufacture in the matured stage of the induction engine gives the impetus to its use for solar water pumping applications in developing countries. IMD pumping water systems have taken a further step towards conventional pumping water systems, with superior sound state switches, high speed manufacturers and highly efficient motor control algorithms being developed. In addition, the IMD fed PV array has been robustly pumped with a VSI (Voltage Source Inverter). The plan discusses a 3-phase IMD to pump solar water that meets the life without the need for electricity in remote areas.

1.2 Literature Review

It offers an examination of device literature from Microgrid(MG), beginning with a collection of sources, topology form (architecture), interconnectedness, customer charge connectivity, operational and control policies and lastly security systems for the MG during normal and defective service. The literature discussed in this chapter is compiled from scientific papers, authoritative foreign journals, and conferences published by scientific and universitarian institutions. The literature in this chapter is described as

- i) Collection of MG services,
- ii) the architecture of MG;
- iii) techniques for optimizing sources for choosing the best power potential
- iv) Methods for service and control
- v) Implementation of the MG system's intelligent metering infrastructure.
- vi) Power converter modelling unit

Solar power plant initial costs are high. Therefore, the emphasis is on achieving the full power from the solar panels of the installed capacity once the plant has been installed. MPPT algorithms involve the development of the developed water pump system directly powered by a PV system for the operation of various irradiation levels and for the extraction of peak power from a solar PV spectrum. Some of these are recommended in MPPT algorithms in. A comparative analysis is given in of the various MPPT technologies. MPPT is an obligatory component of a photovoltaic system from the operational point of view.

The high power of the solar PV array can be obtained using INC (Incremental Conductance) technique. Thus, even with low radiation, the proposed PV pumping system generates peak torque. The INC technology is focused on the analogy of solar photovoltaics with

incremental conductance. In PV water pumping, the main challenge is a prompt regulation of the active energy compared to solar photovoltaic grid interfacing systems.

This is because the engine pump system mechanical time constant is much greater than the above system. The PV array voltage appears to dramatically decrease under sudden drop in solar insolation, and consequently motor streams drop quickly. Once the stream has dropped, the engine begins to draw a higher current that is restricted with the PV array's short-circuit stream to restore the stream. The operating point on the I versus V PV array curves to a current source region is shown by short circuit current and very low voltage. The engine begins running in a stable torque speed zone near a slip point = 1. because of insufficient power.

It risks the engine's health, and whenever an engine reaches the region, a control system needs to be in place to detect this condition and restart the motor from standstill. The engine which frequently enters such situations will reduce the pump's overall job, which is why the MPPT algorithm is responsible for handling such events. A simple IMD control V/f (voltage) connected to the VSI.

The frequency regulatory approach is used in. The pumping system with a DC-DC converter and VSI is used for water pumping applications. However, the solution proposed suffers from the voltage instability of DC connections. V/F is a simple and cost efficient solution. In order to supply power to the centrifugal pump, SAZEPWM technology uses double inverters. The vector control techniques are also complex and they must be supplied with additional current sensors, such as DTC. The V/F power is used to detect only the PV, voltage and DC bus voltage.

By adjusting the modulation frequency to allow the IMD to extract fully, it is monitored by the proposed system energy from the solar panels for various solar insolation levels at a sustainable torque. In comparison with a solar PV powered DC water motor pump, the system is capable of supplying more water. The starting production of the IMD is increased, even when IMD starts with low solar insolation. Even morning-night water is continually pumped.

The induction engine starts with the full current 5 to 6 times, connected to the fixed-voltage AC power supply. Higher solar module numbers are therefore needed to start the engine without any power. However, it is possible to smoothly start an induction motor without drawing high starting current with V/f power, Enhances the engine life. In addition, the grid interfacing PVPs can be used in areas that are blessed by electricity connectivity. There is still no right in Indian sense in many indoor villages and farmland to have electricity.

1.3 Objectives

1. To implement photovoltaic method for water pumping system.
2. To improve reliability and compatibility.
3. To maintain continuity of supply.
4. Increase the motor efficiency.

1.4 Organization of Report

The project explains the implementation of Standalone Photovoltaic Water Pumping System with Reduced Sensors. The Organization of the report is given as follows:

Chapter 2: Deals with the required theory to understand about photovoltaic system and its performance.

Chapter 3: Deals with the solar water pumping system, working and its features.

Chapter 4: Deals with the Simulation, methods and types used in simulation and explanation.

Chapter 5: Deals with the Simulation Results.

Chapter 6: Deals with the advantages, disadvantages and conclusion.

2. PHOTOVOLTAIC SYSTEM

2.1 Photovoltaic water pumping system theory

A PV pumping system consists of five major components, which are PV array, a power conditioning unit, pump-motor load, water tank storage, and pipe distribution system. PV pumping systems can be battery-coupled or directly coupled. The battery-coupled type has PV panels, charge regulator, batteries, pump controller, tank, and DC pump. On the other hand, the directly coupled type has no batteries, and thus water has to be stored in the tank so as to be used at night or on cloudy days. A PV pumping system has two configurations based on the water source. Such a system may be either surface water such as a river or artificial pond or subsurface such as wells.

2.1.1 Motor-pump set in a PV pumping system

Motors are categorized into alternating current (AC) and direct current (DC), where each category has its own types and applications that are matched with the motors' characteristics. The DC permanent magnet (brushed and brushless) and the squirrel cage induction motors are the common types used in PV pumping applications. Selecting the required type of motors relies on the size requirement as well as the type of water source. The motor properties of efficiency, availability, and price are usually considered in selecting the motor. In general, the power demand is the dominant parameter in choosing the appropriate motor. Pumps can be classified into rotodynamic (centrifugal) and volumetric (positive displacement). Each category works at different pumping conditions based on its characteristics. Centrifugal pumps mainly depend on the head, input current, and operating voltage. The pumping is caused due to the pressure difference created between the inlet and the outlet of the pump. The pump's water output increases proportionally with the speed of rotation. They have an optimum efficiency at a certain design head and rotation speed. Centrifugal pumps can be connected directly with PV modules. However, power regulation devices can be connected to enhance their performance such as linear current booster. On the other hand, positive displacement pump's flow rate of water does not depend on the head, but on the speed proportionally. Unlike centrifugal pumps, the match between the positive displacement pumps and the PV array is not suitable. This is because at the beginning of the operation, the motor that is running the pump needs high torque, which means a high starting current.

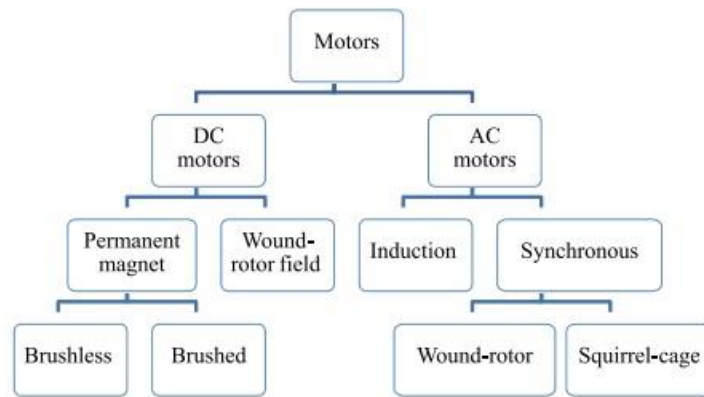


Fig.2.1: Motor classifications chart

2.1.2 Storage system for photovoltaic water pumping system

Storage in PV pumping system means either energy storage or water storage. It is favorable to store water rather than energy due to some reasons such as batteries reduce system's overall efficiency. Moreover, batteries are expensive and need maintenance, which adds more cost to the system. Therefore, water storage is more economical and simpler. The proper selection of a storage tank is important to give the suitable pressure and volume for the used irrigation technique. Moreover, it is also important to design it in an optimal way so as to deliver water for crops without any interruption. It is also vital as the reliability of the PV system is lower than conventional systems.

In addition, plants are irrigated during periods when the solar intensity is low (early morning or late afternoon) as the evapotranspiration is low. Typically, the storage tank is designed to withstand 3 days without any solar radiation to meet the required water demand. The storage tank comes with different configurations. Basically, there are ground, elevated, and hydropneumatic reservoirs. Based on the shape, the elevated reservoir can be classified as a rectangular, an intze, or a cylindrical tank. A cylindrical tank, for example, has features of simplicity in geometry. It is also found that a cylindrical tank is very economical for lower volume capacities.

Regarding the material, there are common materials that the tanks can be made of such as stainless steel, concrete, carbon steel, fiberglass, and plastic. Each material has its own specifications of price, weight, temperature, pressure handling, and corrosion resistance. There are some factors that should be taken into consideration in designing water storage tanks for PV pumping system, such as:

- The tank size should be bigger than the tanks that are used for conventional systems because of the uncertainty of solar radiation. The type of application is important as well. For example, the tank of a PV pumping system for agriculture is usually larger than the tanks used for water supply and livestock.
- The size of the distribution system.
- Available water resources: in some cases, storage tanks are not needed, especially where there is surface water with a year-round flow.

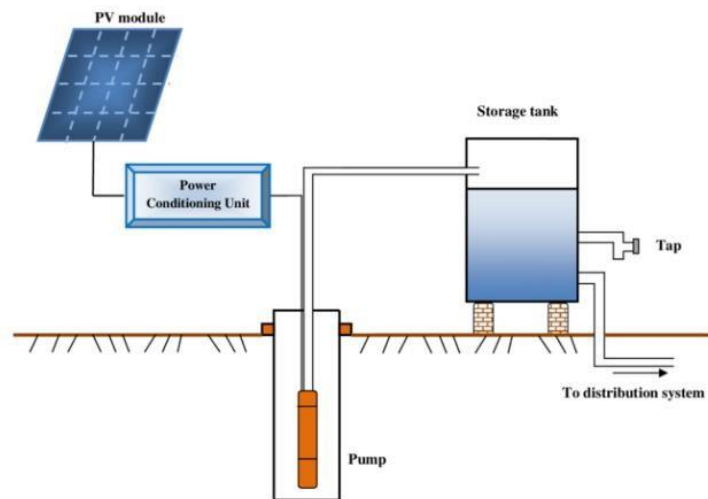


Fig.2.2: Storage of PV water pumping system

2.2 Installation of photovoltaic water pumping system

Many failures of PV pumping systems are caused by pump problems. The PV power supply has much higher reliability than the pump/motor subsystem. A good installation of the pumping hardware will increase reliability. Some things to watch for are described below: Varying water levels: The water level in a well may vary seasonally, daily, or even hourly. The water level in some wells in rocky areas has been reported to drop as much as 75ft during pumping. The pump must be mounted to keep the water inlet below the water level at all times. If the replenishment rate of a well is lower than the maximum possible pumping rate, a level switch or mechanical valve should be included to protect the pump from operating dry. Float switches should be used on storage tanks if the volume of the tank is smaller than the daily pump rate. This will prevent wasting water or pump damage due to overheating.

Protect the pump input: Sand is a primary cause of pump failure. If the well is located where dirt and sand may be pulled into the pump, a sand screen should be used. Most pump manufacturers offer this option or they can recommend methods to limit the risk.

Ground the equipment: Water pumps attract lightning because of the excellent ground they provide. If possible, do not locate the pump system on high ground. Consider erecting lightning rods on higher terrain around the pump. Ground the pump motor, the array frame, all equipment boxes, and one system conductor to the well casing (if metal) or to a bare conductor running down to the water level. Never use the pipe string to the pump as a ground because the ground would be interrupted when maintenance is being performed. The use of MOVs to protect electronics is recommended in areas prone to lightning.

Avoid long pipe runs: When long pipe runs are used in a well with a jack pump, losses may significantly increase. Friction losses depend on pump maintenance. The size of the pipe, the length, the diameter should be larger than the flow rate, and the number of bends pump cylinder. As the output of PV system is limited and varies throughout the day, it is particularly important to keep friction losses low. Pump system efficiency can drop to near zero if a large friction loss occurs. Thus, it is important to limit the friction loss to less than 10% of the head. This can be done by oversizing the pipe, eliminating bends and junctions, and reducing flow rate. Data on pipe size and friction rates are available from pump manufacturers. Pump leathers to be changed by pulling the sucker rod without pulling the pipe.

Protect the control equipment: All electronic control equipment should be housed in weather-resistant boxes. All wires should be approved for outdoor use or installed in a conduit. Any cables used for submersible pumps should be appropriate for that application. Pump manufacturers will give recommended wire types for their equipment.

Use steel pipe: Steel pipe is recommended for use in the well, particularly if submersible pumps are used. Plastic pipe may break. However, plastic pipe provides an inexpensive way to run water from the well to the storage tank or end user.

Protect the well: Use sanitary well seals for all wells. Bury pipes from the wellhead to the tank at a depth that will insure that the pipe won't be broken by traffic or during future trenching or excavation. Mark pipe runs for future reference.

Design consideration of water tanks: Some of the most important factors to consider in designing water tanks follow:

The power source type: The size of the water tank should be greater for PV water pumping systems than for conventional systems because of the intermittent nature of the power source. Poor solar radiation or windless days creates a significant problem in fulfilling the daily water demand. The size of the water tanks for conventional systems depends only on the peak and average daily water demands. However, in the case of PV and wind pumps, the size of the water tanks depends on those factors and local weather conditions.

The geographical location of the system: Geographical location is very important for a PV pumping system. Sunny areas make PV pumps much cheaper and their use ideal in certain locations. Similarly, the size of the water tanks can be smaller in such locations. On the other hand, the size of the water tanks should be much larger for areas of low solar radiation or low wind conditions.

The type of end user: Unlike those in urban areas, PV pumps installed in rural communities are often designed based on minimum water demand requirements.

The type of application: Water reservoirs designed for irrigation purposes should be very large compared to those for other water supply purposes. Water reservoirs designed for cattle watering and community use typically depend on per capita water consumption. As a rule of thumb, there should be 3 days of storage for a community water supply and 5 days of storage for cattle watering.

The size of the water distribution networks: Because they can hold a huge amount of water, large water distribution systems also act as water reservoirs. Small distribution networks with water service at only a few points do not have such options. Many of these services are in rural villages where distribution pipes are usually small. In such cases, it is essential to pump water to holding tanks and then on to distribution networks.

Availability of other water resources: Large tanks may be unnecessary if there is a great amount of annual rainfall in the area. Rainwater can be collected and used for cattle watering and for washing clothes in rural areas. Surface water that flows year round (such as a river) can also provide such services to reduce the need for large-capacity water tanks. Unlike with conventional pumping systems, estimating the proper size of a PV pumping system, including the water tank, requires a detailed evaluation of each component and the system as a whole. As water tanks often cost less than a PV array, installing a larger water tank (instead of a larger PV pumping system) allows for

the possibility of storing water for low solar radiation. Every means to reduce costs must be considered when designing PV pumping systems. For example, for PV pumping systems, using seasonally adjustable tilt angles can increase the daily water production by 5–10%. It is also necessary to supply enough water for days that feature poor solar radiation, and to use the surplus water produced on days with optimum sun. Consecutive days with poor solar radiation are especially difficult when trying to meet daily water demand, but can be offset with a surplus.

Water tanks for conventional supply systems are designed to hold 30%–40% of the total daily water demand. The bare minimum tank size for any community can be estimated by simulating typical 1 year daily weather data and estimating the water balance from the daily water demand and production over the year. The water balance is the difference between the daily water production and water demand.

A negative water balance demonstrates the extra amount of water required over the day. The worst consecutive days of poor solar radiation conditions are reflected on the values of the water balance. The extra amount of water required for these consecutively poor days is calculated by adding the daily water balance of those days with the negative sign that is equal to the amount of extra stored water required to meet the water demand. Because there may be several consecutive poor days, those days with the largest water requirement are exactly the minimum size of the water tank required to fulfill the community's water demand. In the case of hybrid systems, water tanks can be sized like conventional systems because there is a good possibility of getting enough energy to pump water either from wind, sun, or a standby diesel system.

2.3 Photovoltaic water pumping system field performance

As a fact, the performance of any PVPS depends on many factors such as meteorological variables (solar radiation, ambient temperature, wind speed, humidity, and shadow effects), PV module specifications (conversion efficiency and tilt angle), and motor pump–hydraulic system characteristics (I-V characteristic of the motor pump set as well as static, dynamic, and friction loss heads). In this section, PVPS system reliability, feasibility, and field performance are reviewed based on a number of system performances reported in the literature. Eventually, a summary of these results is provided. The latest research articles are reviewed and system performance is reported.

In general, the annual productivity rate (volume of pumping water per 1 kWp) needs to be calculated in order to assess the feasibility of a PVPS.

Also, the PV array, subsystem, and overall system efficiencies, including the performance factor (the ratio of field performance to the theoretical performance), can be used to check the feasibility of a PVPS. In addition, the yield factor (the ratio of the average daily pumped water to the rated pumped water of PVPS) and the capacity factor (the ratio of actual annual pumped water to the annual pumped water of a PVPS that operates at full rating) are other factors that can be utilized to evaluate the productivity of a PVPS.

Moreover, the cost of a water pumping unit is an important assessment factor. the performance of a PV array in a PVPS is evaluated. According to the authors, the field conversion efficiency of the PV array reaches 8.5%. Furthermore, conducted a feasibility study of a PVPS based on a comparison between PVPS and a DG-based water pumping system based on experimental results. According to the results, a large PV water pumping system is required as compared to the DG-based system, which consequently results in a high initial investment.

The efficiency of the available pumping system (motor/centrifugal pump assemblies) is in the range of 35%–65%.

Meanwhile, the overall system performance efficiency is in the range of 3%–10%. The unit cost is found to be about 0.60 USD/m³, which is less than the DG-based pumping system (1 USD/m³). Posorski et al. evaluated 90 PV-water pumping systems installed in seven different countries (Argentina, Jordan, the Philippines, Tunisia, Zimbabwe, Indonesia, and Brazil).

The average overall efficiencies of the systems was 3%. The cost of water discharge is 1 USD/m⁴ and 0.9 USD/m⁴ for PVPS and DG-based water pumping systems, respectively. In the steady-state performance of a PVPS is analyzed. The adopted system consists of a 6.048 kWp PV array, a storage battery with a capacity of 1060Ah, and a centrifugal pump. The switched reluctance motor that is used in the pumping system has an overall efficiency above 85%. According to the results, most of the losses are pumping losses (31.82%) and therefore the efficiency of the pumping system can be improved by selecting an efficient pump. The matching efficiency of the proposed system and the PV array reaches 95%.

These PVPSs have different hourly load demands, PV module types, PV array configurations, storage tank capacities, pumping heads, and pump types. The influence of load profiles is illustrated by a factor called relative deficit (the ratio of the total demand deficit to the total of the demanded water). The results show that there is a linear relationship between the relative deficit and the pumping head. In addition, the relative deficit value depends on the storage tank of the system.

The PVPS with a multistage pump performed the best for a high pumping head, whereas the motor-pump and total efficiencies are 23% and 1.5%, respectively. The studied wells have water depths in the range of 18–75m. Meanwhile, the system water demands are in the range of 36–130m³ /day. The pumping power is in the range of 0.5–1.94 kW.

Four PV water pumping systems installed at four locations in Kairouan in the Tunisia region were evaluated. Each system consists of a PV array with a capacity of 2.1 kW that supplies a three-phase 1.5 kW squirrel cage motor via a 3 kVA three-phase DC/AC inverter. The motor is coupled with an NE 42-30 Pleuger multistage submersible centrifugal pump. A water storage tank capacity of 40m³ volume is utilized in this system. The four systems deliver daily water in the range of 6.5–30m³ /day with a head from 65 to 112m. The maximum subsystem efficiency is 30% at a 65m pumping head, 2.12m³ /h flow rate, and 2840 rpm as the rated speed of the motor. According to the results, the optimal water flow rate is 2.7m³ /h with an overall system efficiency of 2.5. In Ref. [8], the performance of three PVPSs with different PV array configurations was studied.

The performance was investigated based on experimental results to choose a feasible pumping system to provide 4–6m³ per day with a pumping head in the range of 19–35m. The first system (PVPS1) consists of a PV array that has two parallel strings with one module per string. Meanwhile, the second system (PVPS2) contains a PV array with a different configuration (two modules connected in series). The third system (PVPS3) consists of a PV array that has two parallel strings and two PV modules in series in each string. The systems are tested experimentally for water head in the range of 10–35m. The PV array is operated with an efficiency of 12%. The results show that the production of PVPS1 is more than PVPS2 for a pumping head that is less than 30m. Meanwhile, PVPS3 is the best with the aforementioned pumping head values. Furthermore, PVPS1 has the best overall system efficiency (3% at a 25m head) while

for a pumping head that is more than 25m, PVPS2 is better in terms of overall efficiency (3.8% at 35m head).

Four systems with different PV array configurations are adopted in this study. The first configuration (PVPS1) consists of three parallel strings, and each string comprises six 75W modules that are connected in series (6S*3P). The second (PVPS2), third(PVPS3), and fourth(PVPS4) system configurations are 12S2P, 8S3P, and 6S4P, respectively. These systems are tested experimentally for a sunny solar day and based on an 80m water head. According to the results, PVPS3 supplied a maximum daily water volume of 22 m³ and had the maximum average pumping efficiency (45.06%) as compared to other configurations.

hours. The experimental results show that the system has an average daily production of 38m³ with a flow rate in the range of 0.02–0.14m³ / min. According to the results, the overall system efficiency is 2.75%. The system aims to pump water for a 1400m horizontal distance with a 218.34m head.

According to the results, the system starts pumping water at a solar radiation level of 300W/m² while the flow rate was in the range of 1.44–3.24m³ /h. The system is equipped with a 3 kVA/127VDC/AC inverter, a 1.5 kW 127V squirrel cage motor that is driving a centrifugal pump, and a storage tank. The system is modeled and simulated using MATLAB/Simulink. The simulation results show that the overall efficiency of the system is about 3%. Meanwhile, the average daily production is found to be 7.7m³ /day in January and 14.7m³ /day in July.

The system consists of a 1.8 kW photovoltaic array, a SQF5-2 submersible helical pump, and a storage tank. The system is tested with four different heads (50, 60, 70, and 80m). The results illustrate the dependency of the water flow rate and total system efficiency on the solar radiation and pumping head. According to the results, the system with a 50m pumping head has the best total system efficiency (about 7%) at low solar radiation times. With the 80m water head, the highest total system efficiency (around 6.6%) is recorded at medium and high solar radiation only. In addition, the best average efficiency along the operation hours is 4.4% with a pumping head of 80m. According to the results, the cost of the water pumping unit by PVPS is cheaper than the unit pumped by a DG-based system. However, the authors concluded that the cost of a water pumping unit is more sensitive to the price of PV modules than the PV array's lifetime.

Ten different partial shading scenarios are applied to make explicit the effect of shadow on the PVPS performance. According to the results, 50% of the PV production is reduced due to the PV array shadow where the overall system efficiency is degraded from 1.299% (without shading) to 0.83% (with shading). In addition, two shadow cases led to a system shortage. The results show that solar and wind sources are more cost-effective than a DGbased water pumping system.

2.4 Photovoltaic water pumping system design procedures

In fact, the high initial cost of the PV array is one of the important drawbacks of PVPS. Therefore, many researchers are focusing on the optimal size of a PV array as well as other components such as the storage unit and inverter to meet the required load at minimum cost. In general, there are three main methods for sizing PV systems: intuitive, numerical, and analytical.

The simplest one is the intuitive method, which is based on the worst month or the average monthly solar radiation in sizing the PV array and storage units. According to using this method may lead to an over- or under sizing of the PV system, which consequently increases the system cost or decreases the reliability of the system. Thus, the intuitive method is only convenient to be used for estimating initial and rough approximation sizes of a PV system. On the other hand, in the numerical method, hourly meteorological data are used to simulate the system so as to describe its hourly energy flow. After that, based on system performance, possible system configurations that achieve a specific level of reliability are found. Then, an evaluation based on specific constraints is applied to these configuration so as to pick the suitable system.

The main disadvantage of the numerical sizing method is it needs a long computation time to identify one optimal configuration based on a predetermined reliability level. It needs a long time because it computes the reliability of the system for all configurations of design space. In the analytical method, equations for the PV system size in terms of system reliability can be developed and utilized. With this method, the calculation of a system's size is very simple and accurate, but the complexity of deriving these equations' coefficient is the main drawback of this method. Anyway, many issues should be taken into consideration when sizing a PVPS. These issues are accurate information about the water flow rate profile demand, the efficiency and specification of the PVPS components, the total dynamic head, the pumping time factor, the solar

radiation, the PV array power, and the capacity of the storage tank. In this section, research related to PVPS sizing is reported. Moreover, the reviewed methods are summarized at the end of the section so as to provide a clearer comparison. From the literature, the three aforementioned sizing methods are used to design a PVPS. The intuitive methods have been used and applied for many PVPSs. A tool for designing a water pumping system is developed. The design tool aims to match the produced and demanded water. The system is simulated based on specific climatic data. The size of the PV array is then calculated based on the total system efficiency, the solar energy, and the daily hydraulic energy. Similarly, two simplified design procedures, namely peak sun hours and hydraulic energy rule of thumb, are reviewed and compared. The month with the least solar radiation and the greatest water demand is considered as the design month to specify the required size of the PV array. Furthermore, the authors presented a design methodology for a photovoltaic water pumping system. The design procedure starts with the calculation of the solar radiation on tilted PV arrays with an optimal tilted angle to collect the largest amount of solar energy. After that, the optimal sizing of the PV arrays, the inverter, the batteries, the cable, and the motor pump set are done based on the worst month and peak sun hours (PSH).

An economic analysis based on the life cycle cost (LCC) method is carried out for three systems: the PV, the grid, and the DG-based systems. According to the results, the system powered by a PV array was the most cost-effective as compared to the other two conventional power sources. However, 38% of the produced energy by the PV array was not used. a simple sizing method based on standard graphical performance information provided by the manufacturer is proposed to size a PVPS. The authors use a hypothetical constant called an equivalent total head instead of the combination of vertical and head loss. They used DASTPVPS software to experimentally derive an equation that is correlated to the equivalent total head with borehole characteristics. The results showed that the inclusion of an equivalent total head equation boosts the performance of the system and the design accuracy.

On the other hand, the numerical method is used as well for many PVPS. A methodology for sizing a PV array to meet both water and electricity demands is proposed. The methodology consists of two analytical steps. In the first step, experimental data are utilized to compute the area and configuration of the PV array and the maximum capacity of the storage battery of a system that meet the desired energy

demand. On the other hand, in the second step, the authors simulated the proposed configurations to investigate the energy behavior of the system over a specific time period. A PV size versus maximum battery capacity curve is obtained from the numerical algorithm and based on that, the optimal design is chosen. The optimum configuration is selected based on the minimum system cost achieved. A numerical method for sizing a PVPS is presented. The optimization aims to minimize system cost subject to a specific reliability. The method depends on the loss of power supply probability (LPSP) concept. It uses hourly solar radiation and ambient temperature data for 1 year's time. A constant load profile is used in this work with different head levels. Four values of LPSP are used in the simulation: 0, 0.01, 0.05, and 0.1. The simulation program provides several configurations that meet the desired reliability level (LPSP). After that, the optimal configuration that investigates the lowest cost is selected. In, an optimal sizing method for a PVPS based on an LPSP is proposed. The LPSP is used to specify the reliability of a set of system configurations that meet the desired load demand. After that, an economic evaluation is applied to these configurations to find the optimal configuration that achieves the minimum cost at the desired reliability.

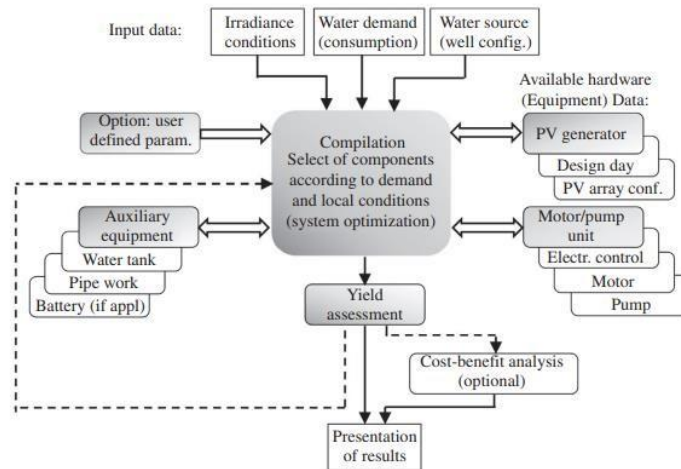


Fig.2.3: Design parameters of a PV-water pumping system

As for the analytical method, it is employed to size a PVPS in many studies as well. The proposed method aims to describe the ratio of PV maximum current to the nominal motor current ($I_{mp}/I_{n.mot}$) as a function of the ratio of PV maximum voltage to the motor nominal voltage ($V_{mp}/V_{n.mot}$) at different values of the LLP. The resulted ($I_{mp}/I_{n.mot}$)-($V_{mp}/V_{n.mot}$) curve divides the plotting plane into two distinct regions called the oversizing region (above the curve) and the undersizing region (below the curve). The motor pump is characterized by I-V and Q-V curves. These curves are

obtained by experiment, and then these curves are fitted by linear and polynomial equations. The method is based on meteorological data, water demand, well depth, simple modeling, and curve fitting techniques. The authors used the speed of pump rotation and the affinity law to find the flow rate of PVPS in terms of speed and head by a two-variable polynomial curve fitting function. After that, the curve fitting method is also used to obtain a correlation for the pump efficiency in terms of the flow rate. The study investigated the effect of total head, peak power of the PV array, water consumption, and storage tank capacity on the performance of the system. The authors proposed a direct relationship between the input power and the flow rate based on experimental data. According to the simulation results, the peak power of the PV array and the total pumping head are strongly related to the relative deficit of the system. On the other hand, the capacity of the storage tank and the profile of water consumption have a minor effect on the relative water deficit with a given total pumping head.

The optimum configuration of the PV array is selected based on an experimental performance of four systems that have 2P2S, 2P1S, 1P2S, and 1P1S configurations that power a 120W/24VDC Shurflo pump. The experimental performance is monitored for three different heads (10m, 15m, and 25m). According to the results, the authors recommended combining the 2P1S and 1P2S PV array configurations together to power the pump. The combination is controlled by an electronic PV array reconfiguration controller to activate the 2P1S PV to power the system in the morning so as to supply the system with an adequate starting current. The 1P2S PV array is used to power the system during optimum sunlight. The approach is represented by optimizing the PV array configuration based on solar radiation. Three PV configurations are used in this research: 4P1S, 2P2S, and 1P4S. These configurations are chosen to address the low, medium, and high solar radiation intensity conditions, respectively. The proposed approach provides a sufficient starting current to the motor, especially in the morning, evening, and cloudy weather. Finally, shows a summary of different techniques proposed for PVPS.

2.5 Photovoltaic water pumping system control procedures

In general, it is important to control a PVPS optimally so as to achieve optimal operation of the system and consequently a reliable system. Many control approaches have been developed by researchers to efficiently operate a PVPS. These approaches include MPPT algorithms, voltage regulation, frequency control, and load matching. MPPT is a technique applied to PV systems to extract and track the maximum power generated by the PV array at different weather conditions. The MPPT algorithms can be classified into conventional algorithms (normally effective in case of not having any shading objectives) and algorithms that are based on stochastic and AI techniques. In general, the conventional algorithms may fail to track the MPP under partial shading conditions and rapidly changing solar radiation. Conventional MPPT algorithms include perturb and observation (P&O), incremental and conductance (In Cond), hill climbing (HC), fractional Voc, fractional Isc, ripple correlation control, and load current or load voltage maximization. The P&O and In Cond algorithms are the most widely used in PV systems. The concept of a P&O algorithm is based on perturbing the output voltage of a PV array with a small value, and then according to the direction of change in PV power, the sign of the next perturbation value is specified. The PV voltage is adjusted based on the comparison between the instantaneous conductance (I/V) and the incremental conductance ($\Delta I/\Delta V$).

A nonlinear control technique that depends on feedback linearization control is used to enable the system to operate at MPPT by adjusting the duty cycle ratio of the buck-boost converter. The concept of feedback input-output linearization control is based on differentiating the PV output voltage to find a direct relationship with a duty cycle ratio. The duty cycle ratio (u) is the control variable that is the output of a nonlinear controller while the solar radiation (G), temperature (T), motor armature current (I_a), PV output voltage (V_P), and output of the linear P-controller (v) are the inputs of the nonlinear controller. Meanwhile, the input of the P-controller is the difference between the PV output voltage and the reference voltage (V_{ref}), which corresponds to the voltage at the maximum power point. Similarly, This method is utilized to track the MPPT of the PV array in a PVPS with minimum losses point tracking (MLPT) for the induction motor. The proposed control strategy improved the system's efficiency and reduced the losses of the motor.

The experimental results indicated that 8% of the input power is saved by reducing the motor losses. An MPPT algorithm and sun-tracking algorithm were applied to improve the overall efficiency of a PVPS that operates with a different total head ranging from 5 to 35m. The MPPT technique proposed is based on the difference between the measured voltage of the PV array and a reference voltage at which the maximum power is obtained. The difference is compared with a sawtooth signal via a comparator to produce the switching signal to control a MOSFET switch. The duty cycle of the triggering signal depends on the output voltage of the PV array and the type of load. On the other hand, the proposed sun tracker for PVPS increased the PV output power by 36%. Two approaches of P&O-based MPPTs are proposed for PVPS. These approaches are the direct duty cycle-based P&O and the reference voltage-based P&O. The effect of the algorithm's Photovoltaic water pumping systems concept 25 parameters (step size and perturbation rate) on the performance of the system is investigated. Furthermore, the drawbacks and advantages of each approach under different weather conditions are demonstrated. According to the results, the reference voltage perturbation has a faster response than the direct duty ratio perturbation at rapidly changing solar radiation and temperature. Meanwhile, the stability characteristics of the direct duty ratio perturbation are better than the reference voltage perturbation technique, particularly with high perturbation.

In addition, the direct duty cycle ratio perturbation technique offers higher energy utilization efficiency than the reference voltage perturbation technique for both slow and rapidly changing solar radiation. In the equation of the terminal voltage of the PV array and the equation of the derivative of power with respect to current at the maximum power point are solved together by a numerical method for maximum voltage and current. After that, a curve-fitting approach is used to describe the current and voltage at the MPP in terms of the photocurrent and PV open circuit voltage, respectively, for all levels of solar radiation. In an electronic circuit is used to produce a fixed duty cycle ratio for the stepup converter to enable the PV array to operate at MPP regardless of solar radiation variations. The results show a large mismatch between the photovoltaic MPP and the equilibrium operating point of the DC motor coupled with a photovoltaic pump. On the other hand, the DC motor-driven centrifugal pump is more compatible with a photovoltaic array.

In matching between a photovoltaic and a three-phase induction motor for optimal operation using an enhanced PV model is presented. A PCU that consists of a double stepup converter and a six-step voltage source inverter is used to track the MPP of the PV array in the system. In an online fuzzy controller is proposed to adjust the duty cycle ratio of a DC-DC converter to optimize the global efficiency of a PVPS by maximizing the input power of a centrifugal pump, and consequently the water pumped flow rate is maximized. The inputs of the proposed fuzzy controller are the variation of both the global efficiency and the chopper ratio. Meanwhile, the output of the fuzzy controller is the adjusted duty ratio. The output and input scaling factors of the controller are chosen based on the trial-and-error method. Five triangle membership functions are chosen to express the output and input variables of the 26 Photovoltaic water pumping systems controller.

3 .SOLAR WATER PUMP SYSTEM COMPONENTS, WORKING AND TECHNIQUES

3.1 System Components

The whole system of solar pumping includes the Solar panels, Voltage Source Inverter, Buck Boost Converter, Permanent Magnet Synchronous Machine. These components are needed for working of the system with high efficiency and to avoid distortions in circuit components like capacitor, diodes and inductors are used in the solar pumping system.

3.1.1 Solar Panels

Solar panels are devices which are used to absorb the sun's rays and convert them into electricity which is given as input to the induction motor. A solar panel is actually a collection of solar cells, which is used to generate electricity through photovoltaic effect. These cells are arranged in a grid-like pattern on the surface of solar panels. Solar panels collect clean renewable energy in the form of sunlight and convert that light into electricity which can then be used to provide power for electrical loads. Solar panels are comprised of several individual solar cells which are themselves composed of layers of silicon, phosphorous (which provides the negative charge), and boron (which provides the positive charge). Solar panels absorb the photons and in doing so initiate an electric current. And the input is given as electricity to the three phase induction motor.



Fig.3.1: Solar Panels

3.1.2 Voltage Source Inverter

A voltage source inverter or VSI is a device that converts unidirectional voltage waveform into a bidirectional voltage waveform, in other words, it is a converter that converts its voltage from DC form to AC form. An ideal voltage source inverter keeps the voltage constant through-out the process. A Voltage Source Inverter usually consists of a DC voltage source, voltage source, a transistor for switching purposes, and one large DC link capacitor. A DC voltage source can be a battery or a dynamo, or a solar cell, a transistor used maybe an IGBT, BJT, MOSFET, GTO. More than one motors can be operated with a single Voltage Source Inverter. VSI is the main component used in the solar pumping system. As, induction motor needs the A.C as input voltage it need to be converted from D.C to A.C.

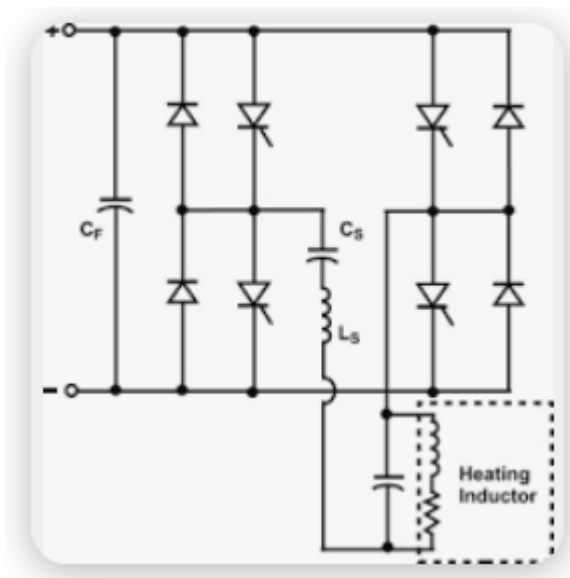


Fig.3.2: Voltage Source Inverter

3.1.3 Buck Boost Converter

The buck-boost converter is a type of DC-to-DC converter that has an output voltage magnitude that is either greater than or less than the input voltage magnitude. Two different topologies are called buck-boost converter. Both of them can produce a range of output voltages, ranging from much larger than the input voltage, down almost to zero. When the buck(step-down) converter is combined with a boost(step-up) converter, the output voltage is typically of the same polarity of the input, and can be lower or higher than the input.

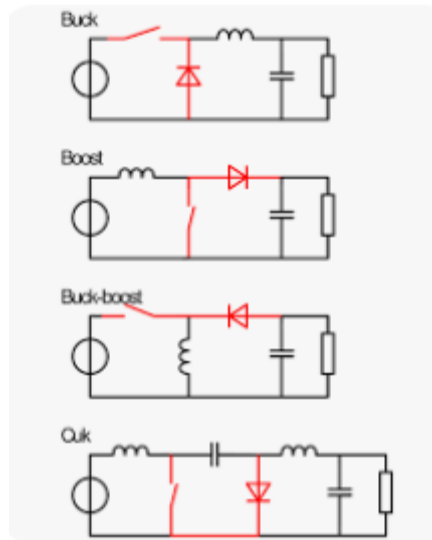


Fig.3.3: Buck-Boost converter

3.1.4 Permanent Magnet Synchronous Machine:

The Permanent Magnet Synchronous Motor (PMSM) is an AC synchronous motor whose field excitation is provided by permanent magnets, and has a sinusoidal back EMF waveform. The PMSM is a cross between an induction motor and brushless DC motor. Like a brushless DC motor, it has a permanent magnet rotor and windings on the stator. However, the stator structure with windings constructed to produce a sinusoidal flux density in the air gap of the machine resembles that of an induction motor. Its power density is higher than induction motors with the same ratings since there is no stator power dedicated to magnetic field production.

With permanent magnets the PMSM can generate torque at zero speed, it requires digitally controlled inverter for operations. PMSM are typically used for high performance and high-efficiency motor drives. High-performance motor control is characterized by smooth rotation over the entire speed range of the motor, full torque control at zero speed, and fast acceleration and deceleration. To achieve such control, vector control techniques are used for PMSM. The vector control techniques are usually also referred to as field-oriented control (FOC). The basic idea of the vector control algorithm is to decompose a stator current into a magnetic field-generating part and a torque-generating part. Both components can be controlled separately after decomposition. The permanent magnet synchronous motors are very efficient, brushless, very fast, safe, and give high dynamic performance when compared to the conventional motors. It produces smooth torque, low noise and mainly used for high-speed

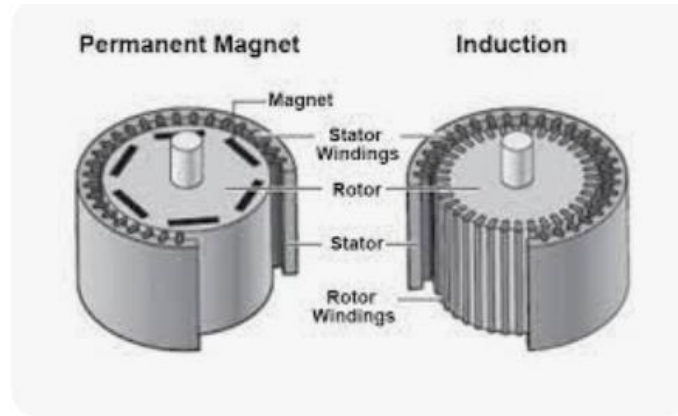


Fig.3.3: Permanent Magnet Synchronous Machine

applications like robotics. It is a 3-phase AC synchronous motor that runs at synchronous speed with the applied AC source. Instead of using winding for the rotor, permanent magnets are mounted to create a rotating magnetic field. As there is no supply of DC source, these types of motors are very simple and less cost. It contains a stator with 3 windings installed on it and a rotor with a permanent magnet mounted to create field poles. The 3-phase input ac supply is given to the stator to start working.

3.2 Solar Pump System Working

A 50-watt photovoltaic solar panel can power a 12-volt pump, which can move 1,300 to 2,600 L/h. Standard plastic fittings and half-inch piping connect these elements to a water saving tank of 500 to 1,000 L. A sturdy stand should be built for the water tank to provide gravity flow, and a frame should also be constructed to provide the best angle for the solar panels.

Multiple filters are needed to protect the life of the pump and minimize clogging in sprinkler emitters and tubes. A solar pump combined with affordable drip irrigation kits can be used with a wide variety of high-value crops to increase water efficiency, minimize fertilizer loss, and irrigate hilly terrains.

In general, the investment required for a PV pumping system is Rs 250-300/Wp (where Rs is the Indian rupee and Wp is watts peak). For example, the cost of a 900 Wp unit would be Rs 225,000-270,000, but with subsidies, this will be reduced to Rs 50,000. To make the best use of solar energy, the PV system, the groundwater pump and the water distribution system have to be well matched.

The PV power provided must cover the power demand of the pump adequately. This is determined by the relationship between the required discharge flow, the total head and the pump efficiency. This depends on the type of pump, which in turn depends on the depth of the available water source.

Although positive displacement pumps are preferred for large heads, centrifugal pumps are most commonly used for this as shown in Figure 1. Photovoltaic (PV) panel electrical outputs are rated according to industry Standard Test Conditions (STC) of 1000 W m^{-2} incident solar radiation at an operating cell temperature of 25°C and under an absolute air mass of 1.5. Environmental conditions met outside the laboratory will cause a decrease in PV performance from the STC rating, the magnitude of which depends on the module technology.

Many additional losses are incurred due to the inefficiencies in transferring energy from the PV panels to a load, such as a pump or battery bank, thus resulting in a secondary decline of performance. Though there have been studies measuring outdoor performance of PV modules, there is a great need for further field studies of complete PV systems.

The advantages of using wireless sensor is to reduce wiring and piping costs, and easier to install and maintain especially over large areas. Energy of pumps used for the agricultural irrigation is generally provided from electrical energy or fossil fuels. Solar energy that is sensitive to environment, clean and requiring no maintenance is an alternative renewable energy source especially for countries like Turkey having a high amount of annual solar irradiation rate.

Means for requirement for irrigation PV pumping systems has advantage of water demand (Anis and Metwally, 1994). In summer months obtained solar energy increases and also naturally water requirement of trees increases.

Solar water pumps are often thought of as being an expensive technology, which is not able to pump enough water and which is not durable. However, solar water pumps have come a long way in 25 years and today there are solar pumps on the market which have improved on previous technology, e.g.: Submersible pumps which can pump up to 200 m heads; pumps that are able to pump larger volumes of water, e.g.: At 100 m, about 10,000 L/day; At 50 m, about 20,000 L/day. Above performance can be doubled through dual systems (if the borehole allows this).

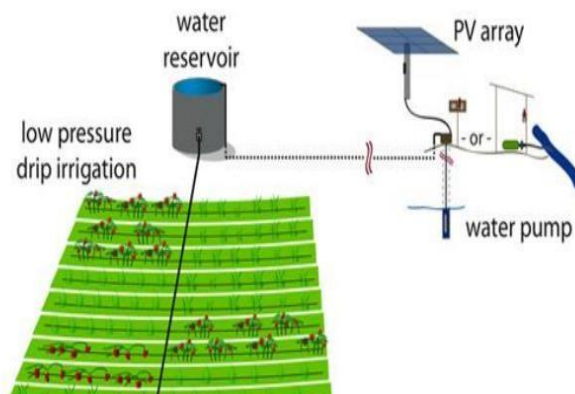


Fig. 3.4: Solar PV Irrigation System

Solar water pumps are often thought of as being an expensive technology, which is not able to pump enough water and which is not durable. However, solar water pumps have come a long way in 25 years and today there are solar pumps on the market which have improved on previous technology, e.g.: Submersible pumps which can pump up to 200 m heads; pumps that are able to pump larger volumes of water, e.g.: At 100 m, about 10,000 L/day; At 50 m, about 20,000 L/day. Above performance can be doubled through dual systems (if the borehole allows this).

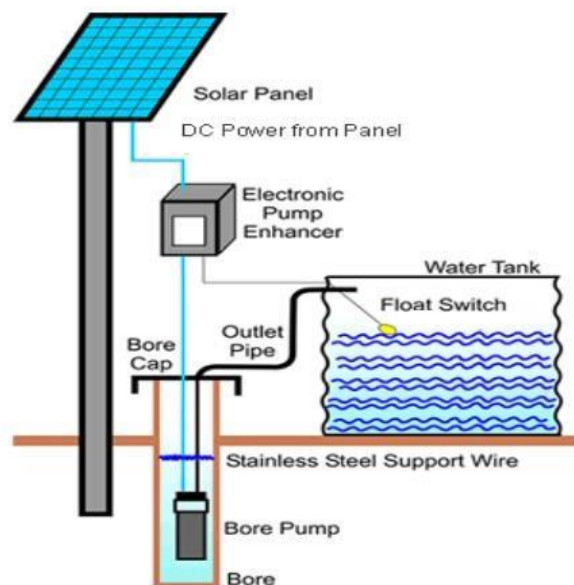


Fig. 3.5: Components of Solar Pumping System

- i) Low maintenance requirements (3 to 5 years)
- ii) Good performance which means fewer solar panels to pump the same amount of water
- iii) Some of the pump models can

be backed-up by a genset to pump additional water with the same pump during the night or during overcast days iv) Good quality and reliability
v) Simple to install.

3.3 MPPT Technique:

A typical solar panel converts only 30 to 40 percent of the incident solar irradiation into electrical energy. Maximum power point tracking technique is used to improve the efficiency of the solar panel. According to Maximum Power Transfer theorem, the power output of a circuit is maximum when the Thevenin impedance of the circuit matches with the load impedance. Hence our problem of tracking the maximum power point reduces to an impedance matching problem. In the source side we are using a boost convertor connected to a solar panel in order to enhance the output voltage so that it can be used for different applications like motor load. By changing the duty cycle of the boost converter appropriately we can match the source impedance with that of the load impedance.

In this we use only one sensor, that is the voltage sensor, to sense the PV array voltage and so the cost of implementation is less and hence easy to implement. The time complexity of this algorithm is very less but on reaching very close to the MPP it doesn't stop at the MPP and keeps on perturbing on both the directions. When this happens the algorithm has reached very close to the MPP and we can set an appropriate error limit or can use a wait function which ends up increasing the time complexity of the algorithm. However the method does not take account of the rapid change of irradiation level (due to which MPPT changes) and considers it as a change in MPP due to perturbation and ends up calculating the wrong MPP. To avoid this problem we can use incremental conductance method.

Promotion of renewable energy technologies is fundamental for economic growth and sustainable development of the country. Renewable energy sources are clean, green, abundantly available and can easily be replenished. Renewable energy sources when utilised for power generation helps to improve system efficiency, reliability and helps to relieve stress on power transmission and distribution infrastructure helping in reducing power deficit. The issues with renewable energy systems used for power generation are intermittent and variable nature of these sources, large area requirement to harness

power, dispatchability, low conversion efficiency and capacity factor. For growth of this sector maximum utilisation of the available renewable source with high conversion efficiency should be targeted. Since renewable sources have non linear power characteristics, there exists one optimum point where power generation is maximum. Locating and tracking this maximum power point is essential for utilising the source to optimum available level and achieve maximum utilisation efficiency. This article discusses the need, methods, advantages of various maximum power point tracking (MPPT) techniques utilised for different renewable energy systems.

Maximum power point tracking for Solar Photovoltaic Systems Need for power tracking in SPV arises because of the rotation and revolution of earth, changing environmental conditions, nonlinear I-V characteristics. Tracking methods are both mechanical and electrical. In mechanical tracking, due to rotation and revolution of earth SPV panels are made to track movement of earth with respect to sun to ensure maximum insolation is received by PV systems. Trackers can be either two-axis, which track sun in both in azimuth and altitude angle so the PV module always points directly to sun or single axis which track only one angle.

Electrical tracking: Power-voltage and Current-voltage characteristics depend upon insolation, temperature, shading, environmental conditions, load connected. Electrical MPP tracking means to make operating point coincide with point where maximum power occurs. Make load characteristics to adjust such that it intersects I-V and P-V characteristics at MPP. Current from SPV module depends on insolation and temperature, I-V characteristics of module will change whenever there is change in Insolation and temperature. This in turn result in change in maximum power point which may not coincide with operating point of load and hence maximum power will not be transferred to load. PV cell have single operating point where value of (I) and (V) of module result in maximum power output. This value corresponds to a particular resistance which is equal to V/I as per ohms law. SPV module has exponential relationship between I and V. Maximum power point occurs at knee of curve where resistance is equal to negative difference of resistance ($V/I = -dV/dI$). A MPPT is power electronic DC-DC converter embedded with algorithm which is inserted between PV module and load to achieve optimum matching. Purpose of MPPT system is to sample the output of PV cells and apply proper resistance (load) to obtain maximum power for

any given environmental condition. To SPV module DC to DC converter acts as load and hence input resistance of converter is of interest. In R_{in} of converter lies on V_{mp} - I_{mp} point, maximum power can be transferred to the converter and in turn to load.

4. SIMULATION AND EXPLANATION

4.1. Introduction

For the implementation of a Simulink course, this archive is important. This course is for people who have never used Simulink. The class is made up of two sections. There are activities in another repository that take you on the tasks required to produce and use a Simulink model. When you start using Simulink, you will find a lot of utility that is normal to yourself. There are inevitably aspects which need to be explained a little.

Thus, a part of the seminar is a debate and a demonstration here. This record includes the discussion notes. All of Simulink's work is hard to bring into a particularly short article, so I've concentrated on the package bits that I think are the best. I've also tried to provide functionality that the easy consumer cannot see. The intention is for you to use these notes as a guide for the work and the design of your own models.

Although these notes have limitations, I hope it should be sufficient to start using the package and cover most of your demonstration needs. It's not a manual for the Simulink. You can eventually know more about everything inside Simulink. The documentation included with in the package is intended to be used in this folder. Math works provide Simulink with comprehensive online documentation that can be used with the support system of the MATLAB.

There is such an abundance of online documentation that very few individuals have the ability or tendency to peruse every last bit of it. Therefore, the aim of the seminar is to emphasize things you should think about Simulink and to think about where you might find any other data you like.

4.1.1. The Solver

You can only run your model most of the time using the default settings. However, often you want to use smaller steps or fixed width steps in the model. All of this can be configured on the Settings Solver tab. Choose Solver from the menu bar on your model.

4.1.2 Simulation

Model Configuration Parameters

Then pick Solver from the menu on the left.

Simulation time

The time box for simulation is located at the top right. I recommend you keep it as zero for the beginning period. The time of the pause is the same as the time at the top of your model on the icon bar.

Solver options

Two solver forms are available. A phase solver variable is used by default. This sets the phase size automatically as the model runs. I suggest you retain the default solver if you are using variable phase (ode45). Set a fixed value of the Max step to boost the smoothness of all graphs if necessary. For models with distinct components a move to a fixed width solver is necessary. If there is no continuing component, shift solver to discrete (no continuous states). I suggest you also set a known step size value.

In order of simplification, the fixed solvers are numbered. Click the help button at the bottom of the parameter set up window to get more information on solvers when opening the solver portion. The simplest thing is Ode1. Select Selecting a Solver on the bottom of the list.

Zero-crossing options

The zero-crossing options box is available at the bottom of this tab. If you think it helps, you can deactivate a zero crossing control.

4.3 Sources Library

The Sine Wave Block

In the source library, most blocks extend themselves. It is easy to use the fundamental sin wave block. In block parameters, you just set frequency and amplitude.

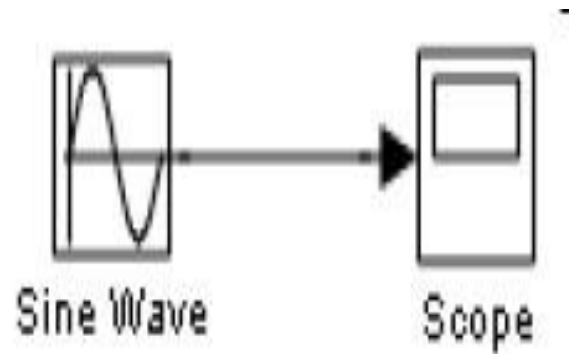


Fig. 4.1: Sine Wave Block

During exercises, we use the Sine Wave block to measure the response of the device. It is tedious, however, to constantly edit block parameters to change the sine wave frequency. An additional time input to the block is used as an alternative.

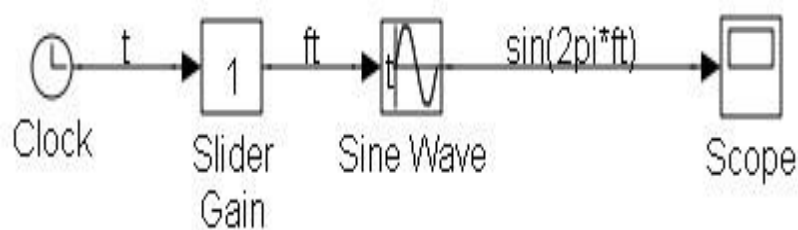


Fig. 4.2: Block Diagram of Sine Wave Block

In the Sine Wave block parameters you can set the time parameter for use with an external message and set the frequency to 2π . A slider gain will then attach a clock to the new input. Then the slider sets the sine wave frequency.

From Work Space Block

This block is used to import data from the MATLAB workspace into the Simulink model.

One simple approach is to import the data, as shown below, into the matrix.

time	Signal 1	Signal 2
0	1	1
5	2	4
10	3	9
15	4	16
20	5	25

The variable name of the matrix is entered in the parameters of the block. Notice that the above data are very simple in the matrix. In order to compute values between the given points at any time, Simulink will use interpolation. In a structure, the block reads also in data. The following MATLAB script will construct a Simulink-readable structure. To use Mydata in a workspace block's data parameter.

```
%Generate data to go into simulink
%Generate time vector
n = 21;
t = linspace(0,10,n);
t = t'; %transpose into a column
%Generate two sets of data
x = rand(n,1);
y = t.^2-50;
%Write data into structure
Mydata.time = t; %Time vector
%Signal
Mydata.signals.values = [x y];
Mydata.signals.dimensions = 2;
```

Fig. 4.4: MATLAB script for Simulink-readable structure

4.4 Scope Block

Axes Properties

Click on the axes right, then choose Axes. This enables the axis and title axis limits to be changed. The default axis title is the mark that shows the signal. It is therefore possibly easier to mark the signal than to send the title here directly.

Scope Icons			
	Print		Magnify in x and y
	Parameters See below		Magnify in x only
	Auto Scale		Magnify in y only

Fig.4.5: Axes Properties

Parameters / general

To pick General tab, click on icon Parameters. You can then set the number of parameters.

The graph's axis and time range.

Parameters / Data History

Click the Settings button, then click the Information tab of History. The scope data can be saved into the MATLAB working area and the size of stored data can be limited. You may store the data in a frame or array. In a structure, the following function shows the details. You can also draw data from a Workspace Block by using the function.

```

function PlotScopeData(ScopeData)
%Plot graphs of data saved by a scope block
n = length(ScopeData.signals); %Number of axes
t = ScopeData.time; %time vector
for k = 1:n
subplot(n,1,k); %One subplot for each axes
x = ScopeData.signals(k).values;
plot(t,x);
xlabel('time');
ylabel(ScopeData.signals(k).label);
title(ScopeData.signals(k).title);
end

```

Fig. 4.6: MATLAB Script for Saving Data

Parameters / Style

This allows the colour and markers for the data to be changed.

4.5 Math Library

Sum, Add and Subtract Blocks

The add, subtract and sum blocks are practically similar. You can convert one to the other by adjusting the icon shape and sign list of the block parameters. The diagram at the right provides examples of how the list of signs and the shape of an icon are modified.

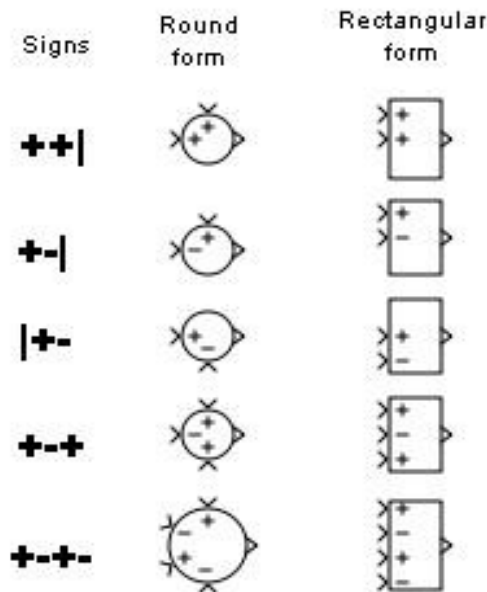


Fig. 4.7: Add and Sub Blocks Representation

Product and Divide blocks

Also interchangeable is the commodity and separating blocks. You can set the number and distribution of inputs using List of slashes in the block input number and slashes forward. For example, by inserting `/*/**` in the input number parameter, a block to the right was generated.

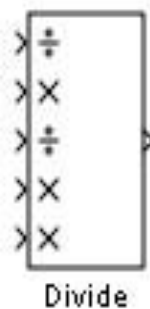


Fig. 4.8: Product and Divide Block Representation

4.6 Mathematical Functions

For various mathematical functions there are several blocks. The Abs block recognizes the signal size or value, the Unary minus block excludes a signal and attaches a signal constant to the Bias block. In combination with the signal (positive or negative) of the input, the sign block produces a plus or minus one.

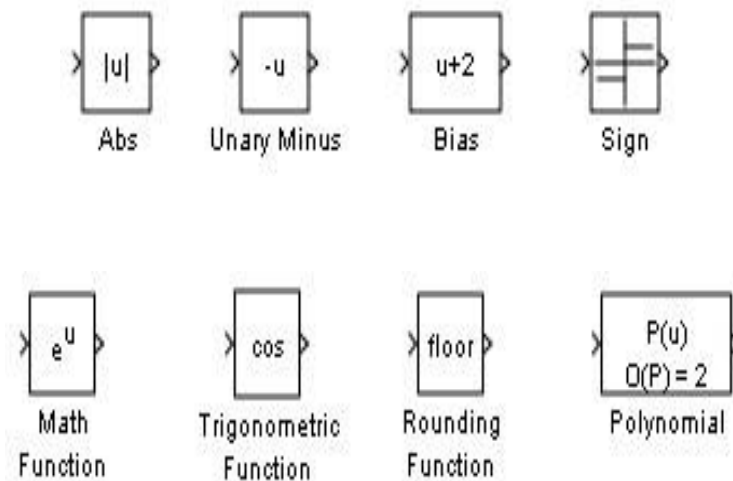


Fig. 4.9: Representation of Mathematical Functions

The Block Math Function can fulfil various functions, including cube, square, log, mutual etc. You can pick the specific function you want by using a block parameter. If you want a trigonometric function, the trigonometric function block works in the same way. The rounding function block completes an integer. There are four various MATLAB functions available, such as floor, ceiling, round and ceil. The feature you want is picked. See support for the MATLAB features for more info.

You may use the Polynomial Block to evaluate a polynomial with the input as an argument. Polynomial coefficients are entered in a block parameter as a vector. The MATLAB function flexibility is similar.

Complex numbers

Signals can be complex in Simulink if needed. If it is possible to work with a block. It'll most likely be complex numbers. There are special blocks for the development of complex numbers.

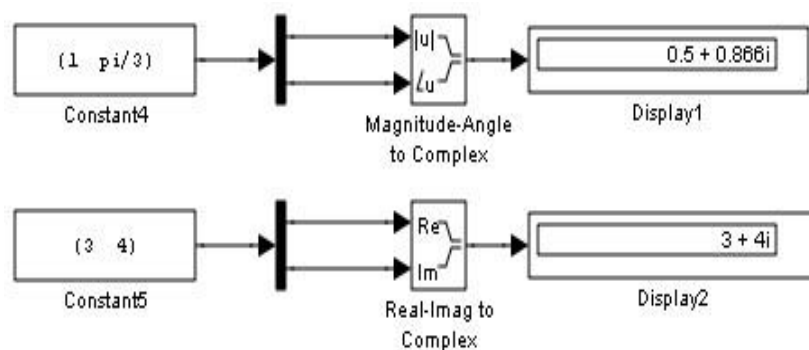


Fig. 4.10: Representation of Complex Numbers

Blocks may also be derived from the different components of a complex number.

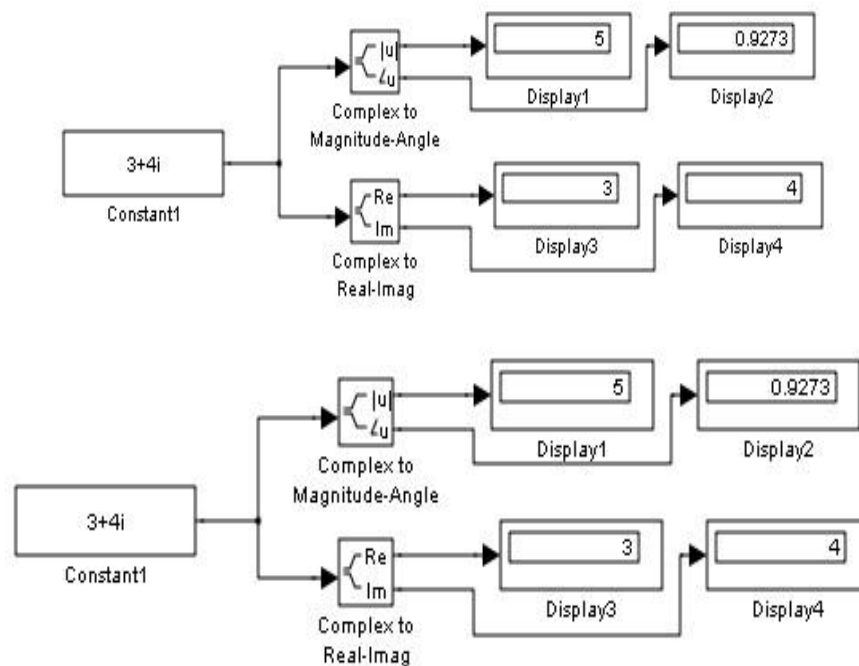


Fig. 4.11: Different Components of Complex Numbers

Sadly, integrators don't work with complicated inputs.

Matrices

Since Simulink is based on MATLAB, the matrices and multi-dimensional arrays are often used. It is no surprise. For example, the basic gain block can be set for the matrix or array multiplication. Special blocks for matrices are also available. You should use the assignment block. Substitute another signal in a specific element.

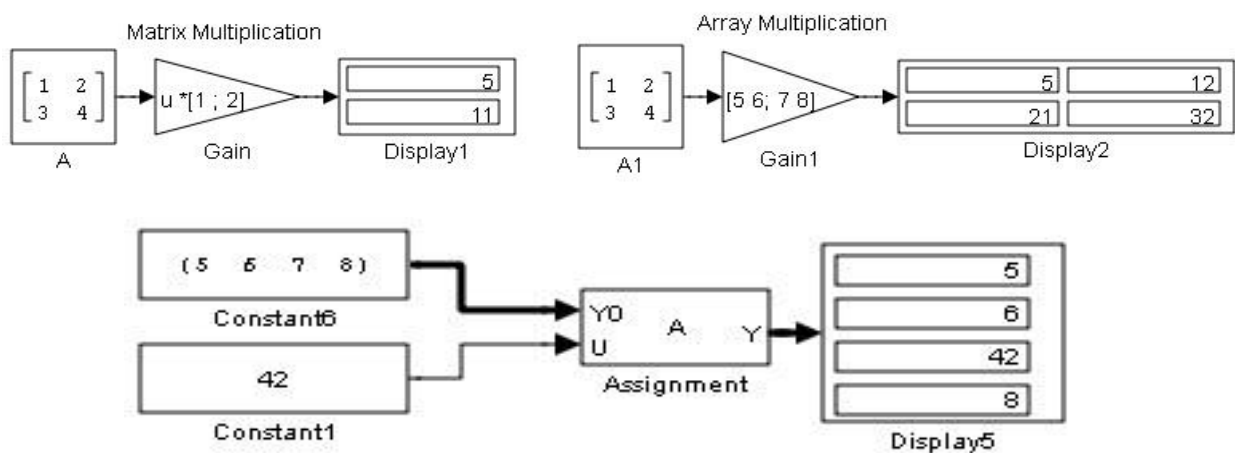


Fig. 4.12: Multiplication in different cases

In this case, the 3rd variable in Y0 should be superseded by the input U parameter.

Note that a column vector will be generated, even if a row vector is entered, by default when entering a block vector. The way to solve the problem is to uncook the vector parameter 1-D constantly.

Permit the size of the block exchange. 2D is like transposing, but can work with many arrays as well. Concatenate can also be used to connect lines (size 1) or columns (size 2) to the matrix. Blocks are also available for vector element operations.

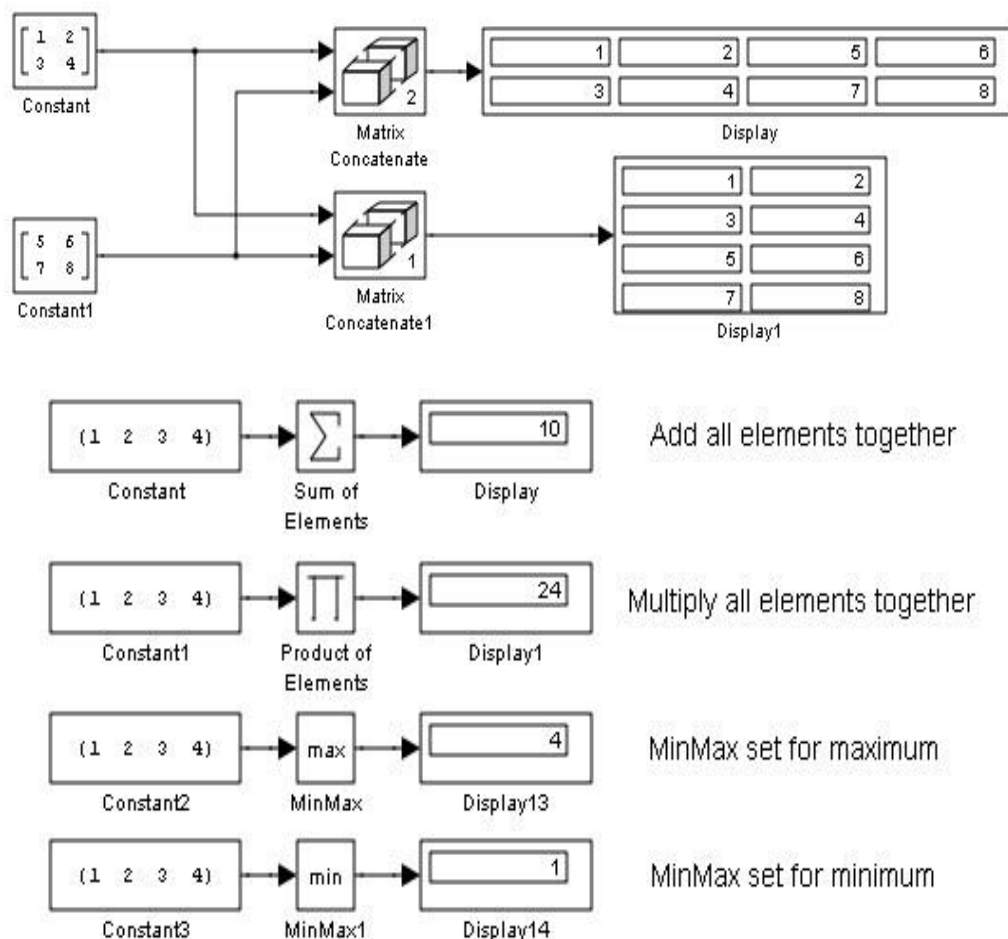


Fig. 4.13: Vector Element Operations

4.7 User Defined Functions

The exact role you want cannot be identified often. If so, then I would advise you to check out the block Function first. You can then use a restricted set of operators and functions to type a mathematical term.

Click the Aid in the parameters of the block to see what is allowed. The MATLAB fcnblock you can write your own MATLAB function, if you need anything more complicated.

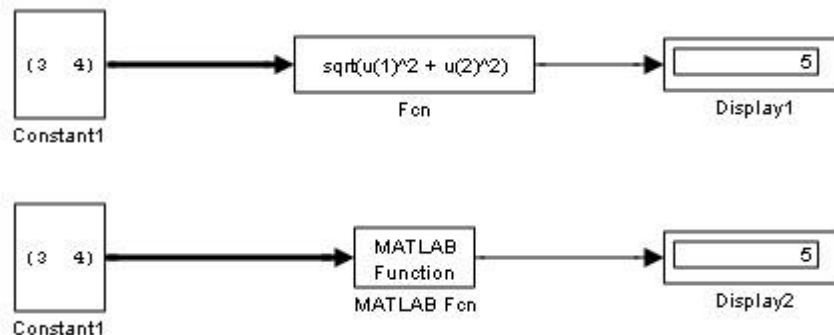


Fig. 4.14: MATLAB User Defined Function Block

The MATLAB function myabs below has this block configured to call. x is input, y is input the output.

```
function y = myabs(x)
a = x.^2;
b = sum(a);
y = sqrt(b);
```

You have to use the Embedded MATLAB Function or S Function if you use Simulink to construct a program using the Real Time Workshop. S functions are used in your model to combine MATLAB, C and Fortran.

4.8 Look up Tables

If anything else goes wrong, you can set your role by using a search table.

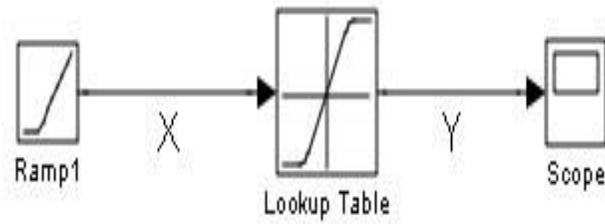


Fig. 4.15: Look Up Table Representation

This must be achieved by two vectors. A input vector and a vector table. Corresponding performance values. The parameter in the block is defined. The output is measured with interpolation for intermediate values. For the estimation of values outside the same input set, extrapolation is used.

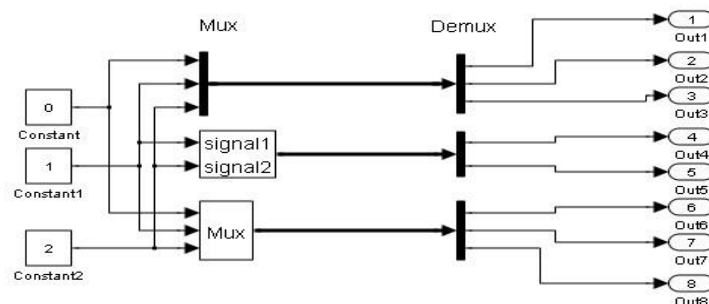
4.9 Signal Routing

Mux and Bus

A mux is a multi-signal cable. Use the mux block to construct a mux. All three blocks with different display choices under the phrase "Mux" are mux blocks. Signals must have the same form and scale inside a mux. You can feed a mux directly on aMathematical block to work on all the signals in the mux.

The individual wires in a mux are separated by a demux block. Normally a single integer includes the number of block outputs in the demux block parameter. When entering a vector in its output number, each number in the vector determines the number of signals in a given output. In [1 3 1], for example, three outputs are defined. Single wires are the first and last effects.

As in the diagram below, the middle output is a mux with three signals.



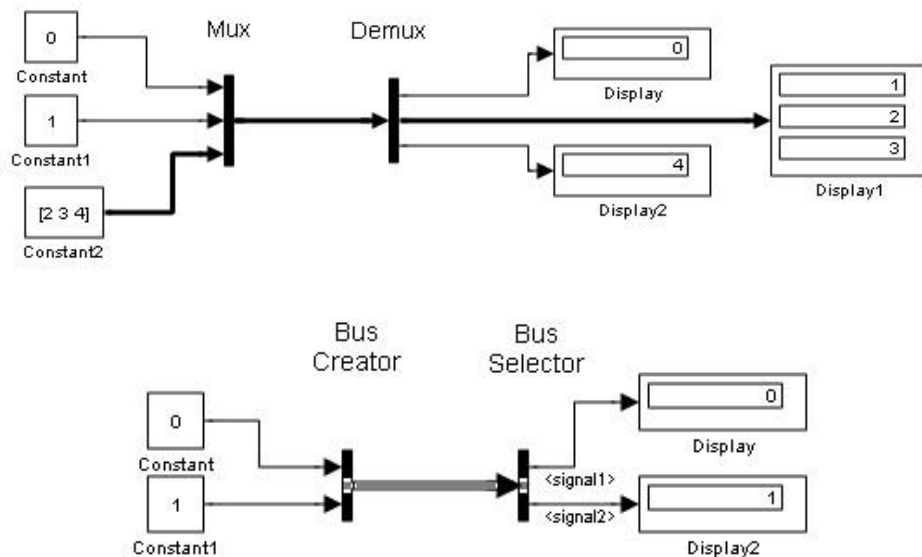


Fig. 4.16: MUX and BUS Operations

A bus in a vehicle is like a cable car. A convenient way to keep things secure and not have a nest of birds with cables that run around. It may contain various types and sizes of signals. This means a mathematical block can not be fed by a bus. A bus and mux can be nearly interchanged unless the bus comprises numerous signal types.

Selectors and switches

The selectors are the MATLAB subscription simulink equivalent to extract 3rd and 6th element of the input vectors, the selector above was configured. There were 6 inputs, while [3 6] was indexed. Selectors may also be modified to get the indexes from an additional entry. Selectors are still working with higher dimensional arrays and matrices.

There are three types of switches. In the center pin is calculated the output of the switch block. The central input is compared to an internal threshold value. You may pick a bigger, smaller, equivalent contrast, etc. If the relationship is valid, the output is updated to the highest input and the output to the lowest input is associated.

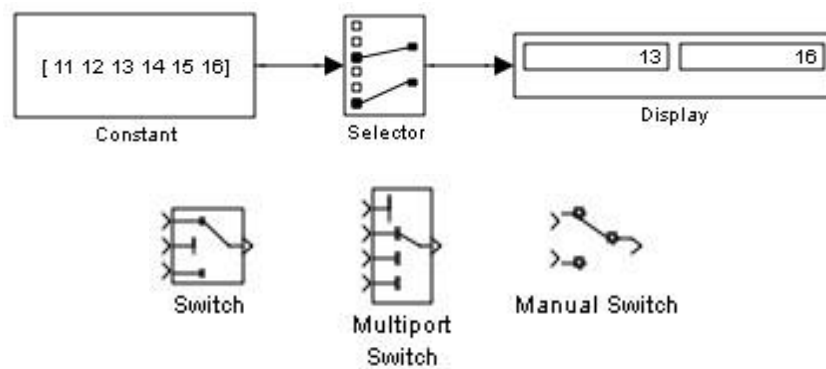


Fig. 4.17: Selectors and Types of Switches

An integer at the top of the Multipoint Switch Input shows the other inputs that are connected to the output. When the highest input is, the first of these inputs is picked. The second vector index changer is similar technically to a multi-port switch if the top two are selected, etc.

Double click on the button toggles a manual switch.

4.10 Sub Systems

The MATLAB script is a subsystem. A group of blocks for the execution of a specific mission. By dragging the subsystem block from the Ports and Subsystems library to build a subsystem from scratch or selecting blocks you want to place in the subsystems, you can pick them

Diagram ► Subsystems & Model Reference ► Create Subsystem from Selection

The menu bar of the model. You can find them in the Ports and Sub-Systems library if you need additional input and output ports. The source and sink library is also the required source. 1. Enabled Subsystems: Input to your subsystem may also be added. Drag the permission block to your subsystem from the Ports and Subsystems library. A subsystem on which the input is allowed is disabled by a zero. You can customize the block settings of the output ports to retain the present value, or to reset the value when the subsystem is disabled to a specified value.

2. Triggered Subsystems: In your subsystem you can also add an input trigger. Whenever the signal needed for the trigger entry is applied, the subsystem will then perform one phase. The triggering input can be configured to respond to a rising edge, a falling edge or both.

5. SIMULATION RESULTS

5.1 Waveforms

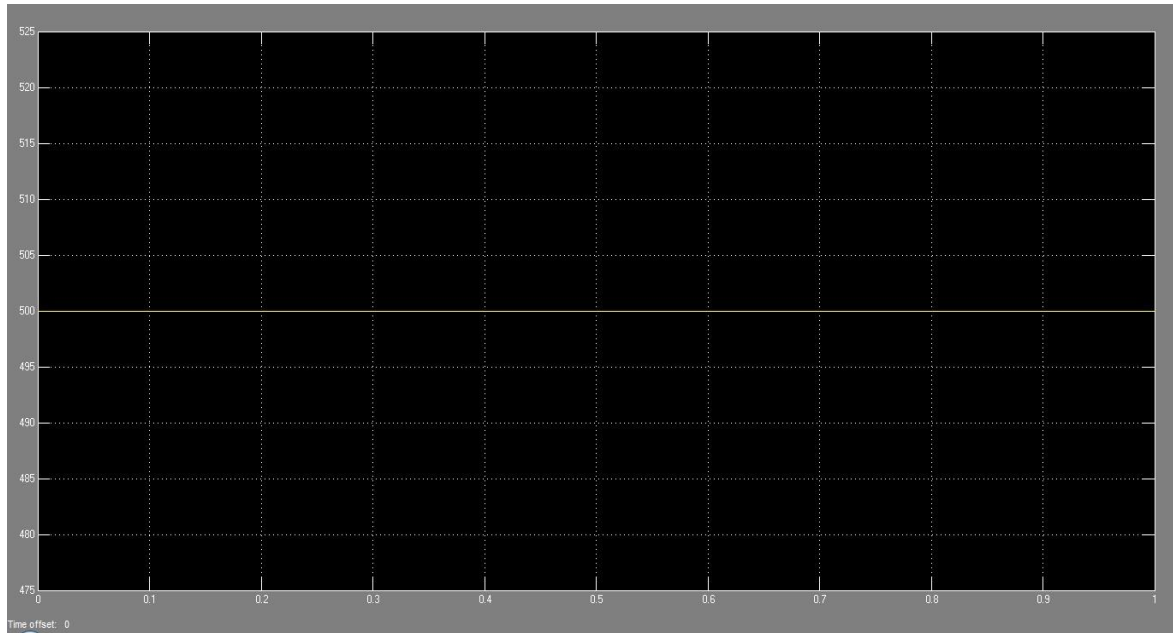


Fig. 5.1: Irradiation Graph (Input to the Solar Generation System)

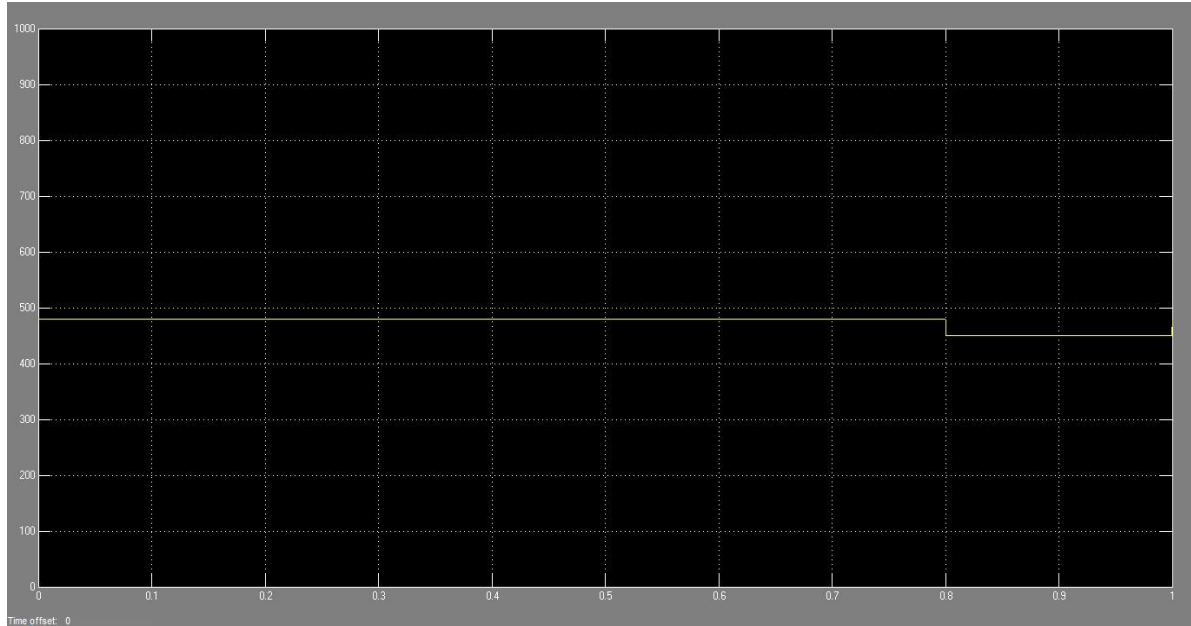


Fig. 5.2: PV (Solar) Voltage Waveform which is coming from PV

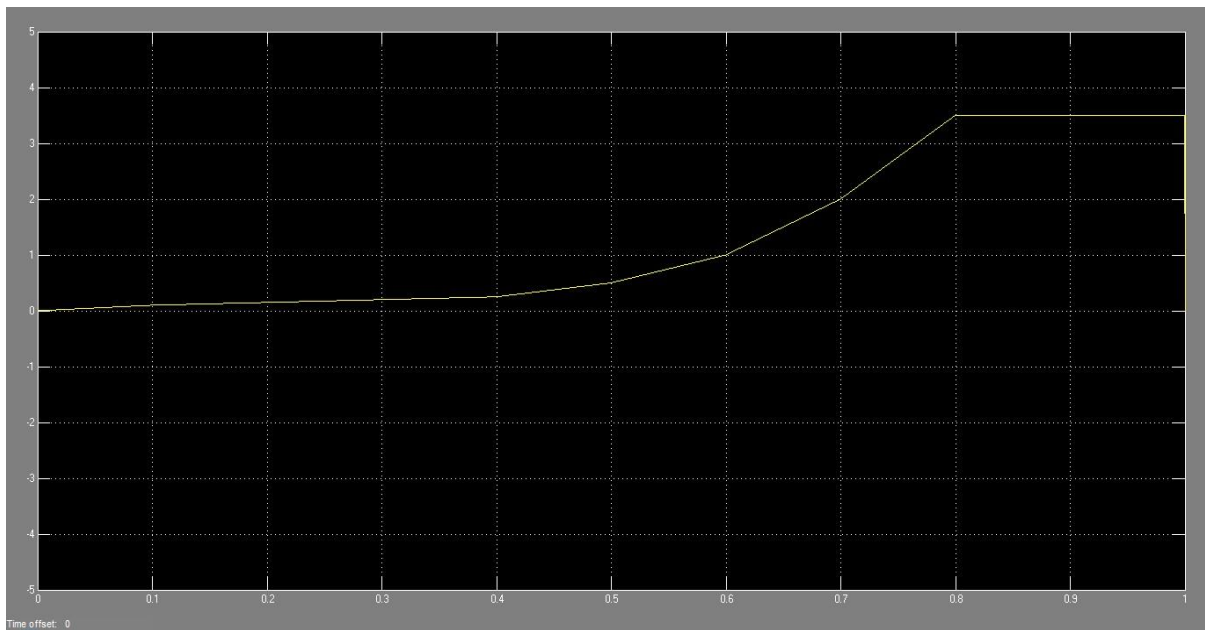


Fig. 5.3: PV (Solar) Current Waveform which is coming from PV

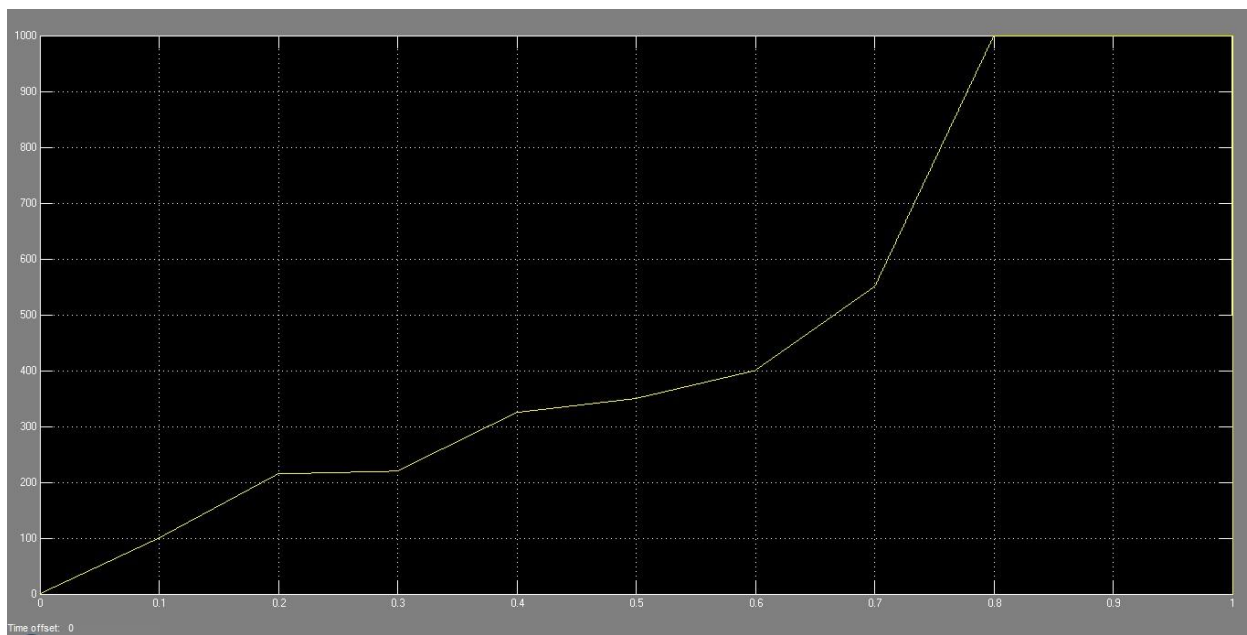


Fig. 5.4: PV (Solar) Power Waveform which is coming from PV

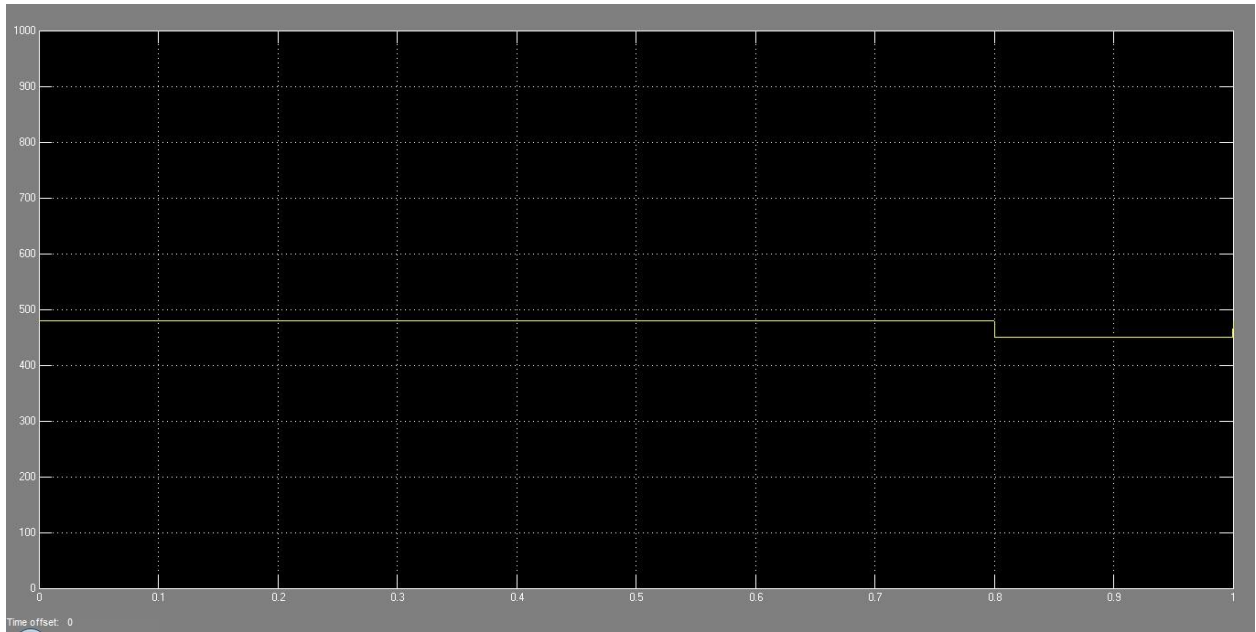


Fig. 5.5: DC Link Voltage Waveform (Output of PV and Input of Three Phase Inverter)

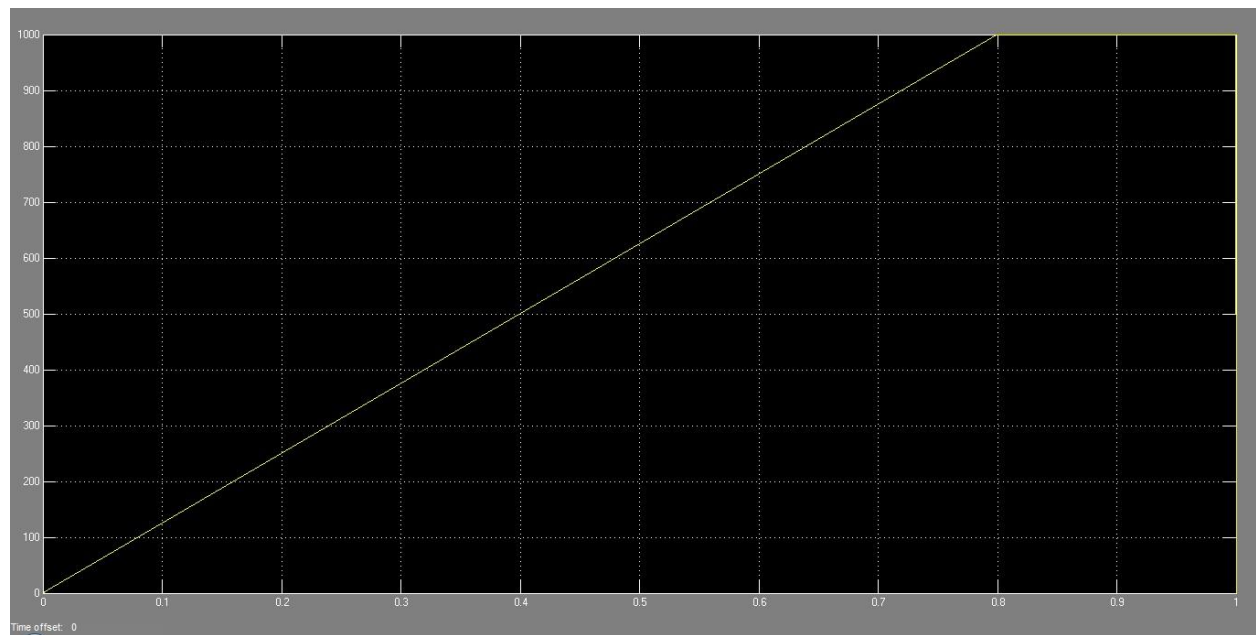


Fig. 5.6: Speed of a Motor

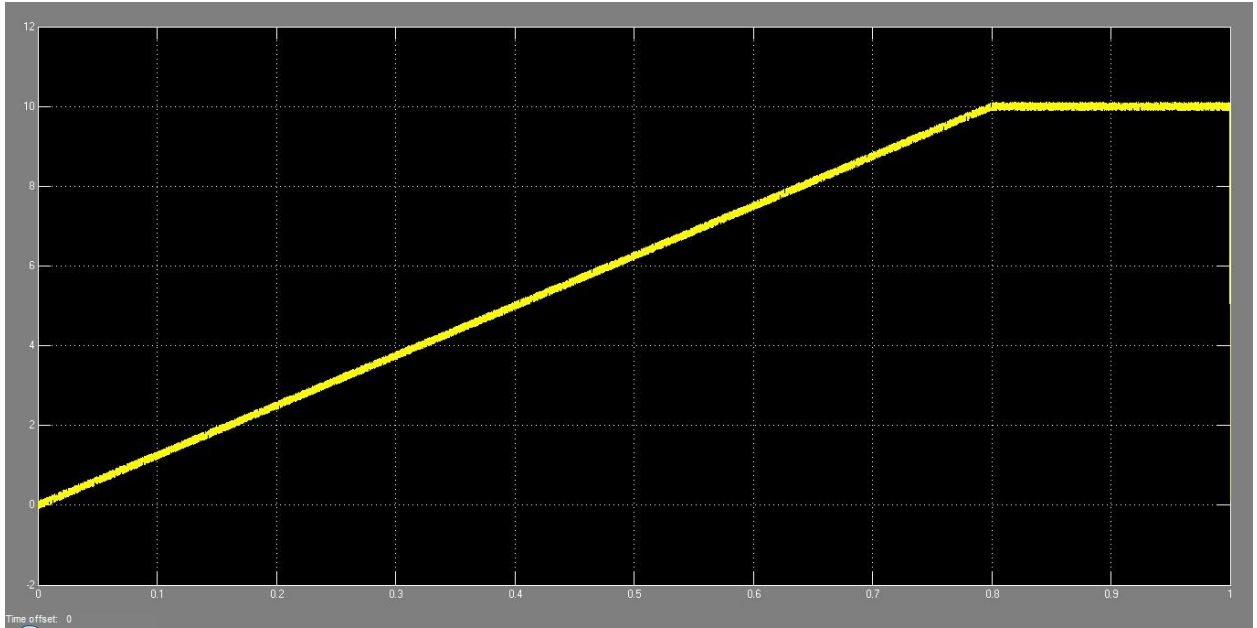


Fig. 5.7: Electro Magnetic Torque

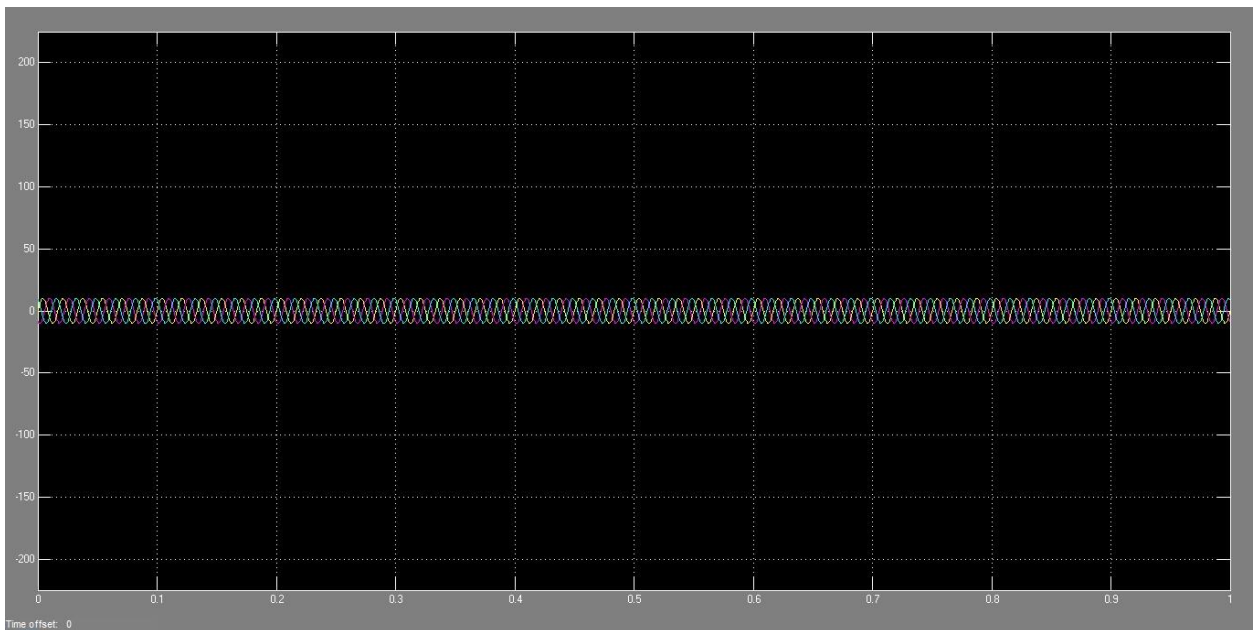


Fig. 5.8: Three Phase Motor Current (Output of the Inverter)

5.2 Result

A standalone solar PV system has been modelled for the induction motor drive used in water pumping system. Solar PV system fed induction motor via series integration circuit of boost converter with constant voltage control and inverter has been presented. Simulation results shows that induction motor can work effectively by PV System and there are some unwanted disturbances to be eliminated, a better performance in terms of the time of response pumped water; flux ripples and the stator currents are reduced

6. ADVANTAGES, DISADVANTAGES

6.1 Advantages

1. The proposed system tracks the MPP point by altering the modulation frequency so that the IMD is able to extract the maximum power from the solar PV array at sustained torque for different solar insolation levels.
2. The proposed system is able to supply more water as compared to a solar PV fed DC motor based water pump.
3. This also improves the life of the motor.
4. As standalone is used, energy is stored at night time without usage of any battery.
5. There is no need of power, as the solar panels acts as the input to the induction motor.

6.2 Disadvantages

1. The initial cost of solar power plant is high.
2. The fact that the mechanical time constant of the motor pump system is much higher than that of aforementioned system.
3. Under sudden fall in solar insolation, the PV array voltage tends to reduce drastically and consequently the level of flux in the motor falls rapidly.

7. CONCLUSION

The independent, reduced sensor, photovoltaic water pumping system was proposed. a standalone photovoltaic water pumping system without battery storage is investigated. The proposed system consists of using a variable step size incremental conductance method to adjust the duty ratio and fuzzy logic control based on DTC to control the induction motor Just three sensors are used. The reference rate generation for V/f control system has been proposed based on the available power that controls the active power of a DC bus. The flux and torque are controlled separately. For speed regulation in an induction motor drive, the PWM frequency and the pump affinity regulations were utilized. Its operational viability was confirmed by simulation and testing. Many performance conditions such as start, radiation variance and stability have been checked and fulfilled experimentally. The main contribution of the proposed control system is that it is fundamentally excluded from the constant approximation error of the pump. Also with different radiation, the machine monitors the MPP with an appropriate tolerance. In industrial and irrigation fields these solar water pumps helps in usage of water without using power.

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